24，求序列的DTFT

因为matlab没有内置DTFT的函数，因此用定义$X(k)=\sum_{n=-\infty}^{\infty}x(n)e^{-jwn}$来求解析解

```
syms w n a w0;
f1=exp(-1i*w*n)/2^n;
sum1=symsum(f1,n,-3,-1)+symsum(f1,n,0,inf);
pretty(sum1);

f2=a^n*sin(n*w0)*exp(-1i*w*n);
sum2=symsum(f2,n,0,inf);
pretty (sum2);

f3=exp(-1i*w*n)/4^n;
sum3=symsum(f3,n,0,inf);
pretty(sum3);
```

分别对应三小问，输出为

```
{                                                                      1
{                          Inf + #3 + #2                      if  #1 == -
{                                                                      2
{
{                     / /          / 1 \n              \     \
{                  #1 | |     lim  | - |  exp(-n w 1i) | - 1 | 2
{                     \ \ n -> Inf \ 2 /               /     /         1
{ 2 #1 + #3 + #2 - ---------------------------------------------  if  #1 ~= -
{                                     2 #1 - 1                         2

where

   #1 == exp(w 1i)

   #2 == 8 exp(w 3i)

   #3 == 4 exp(w 2i)
```

```
{  1i Inf + #2   if  #3 == 1 and #4 ~= 1
{
{ - 1i Inf - #1  if  #4 == 1 and #3 ~= 1
{
{    #2 - #1     if  #3 ~= 1 and #4 ~= 1

where

   #1 ==

     /                                              /           n                              \ \
     | exp(w 1i) exp(w0 1i) - exp(w 1i) exp(w0 1i) |     lim    a  exp(-n w 1i) exp(-n w0 1i) | | 1i
     \                                              \ n -> Inf                                 / /
     ---------------------------------------------------------------------------------------------
                                     2 (a - exp(w 1i) exp(w0 1i))

                / /          n                            \     \
         exp(w 1i) | |     lim   a  exp(-n w 1i) exp(n w0 1i) | - 1 | 1i
                \ \ n -> Inf                              /     /
   #2 == ---------------------------------------------------------------
                          2 (exp(w 1i) - a exp(w0 1i))

   #3 == a exp(-w 1i) exp(-w0 1i)

   #4 == a exp(-w 1i) exp(w0 1i)
```

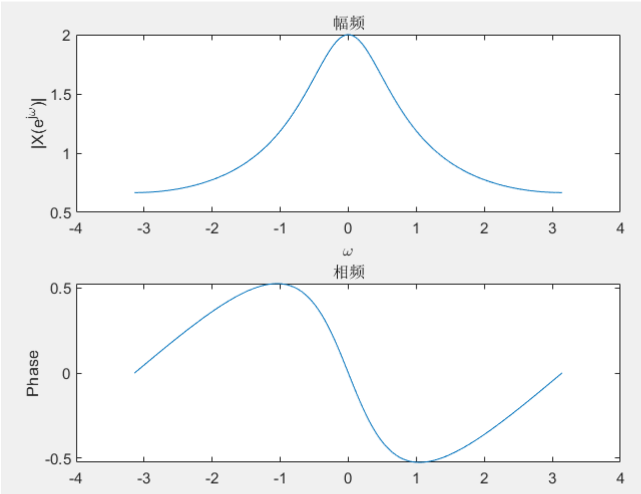
```
{                                                                   1
{                              Inf                     if  exp(w 1i) == -
{                                                                   4
{
{              / /          / 1 \n              \     \
{   exp(w 1i) | |     lim   | - |  exp(-n w 1i) | - 1 | 4
{              \ \ n -> Inf \ 4 /               /     /             1
{ - -----------------------------------------------------  if  exp(w 1i) ~= -
{                       4 exp(w 1i) - 1                             4
```

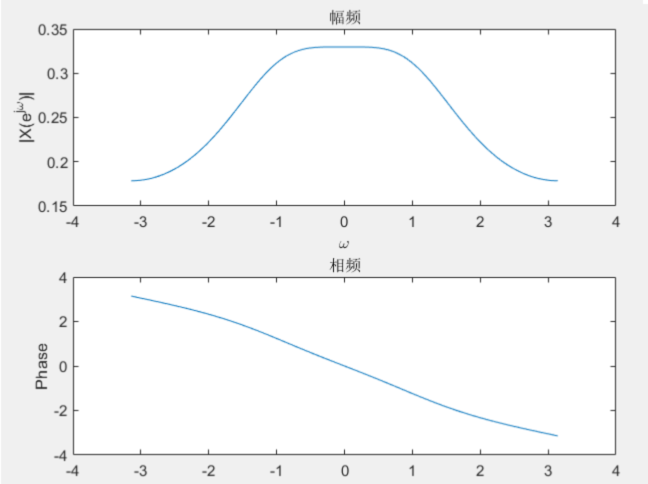
但是离散信号DTFT的解析解太抽象了，画出幅频和相频图象

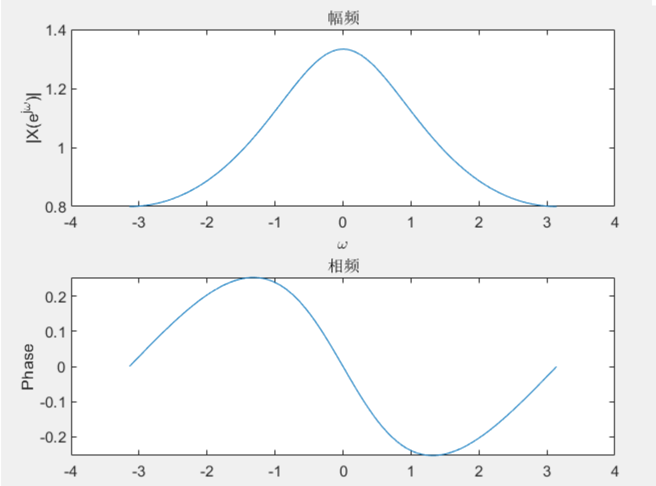
```
N=100;
n=0:N;
a=0.3;
w0=1;
x1=(1/2).^n;
x2=(a.^n).*sin(w0.*n);
x3=(1/4).^n;
w=linspace(-pi,pi,1024);
X1=zeros(1,length(w));
X2=zeros(1,length(w));
X3=zeros(1,length(w));
for k=1:length(w)
    X1(k)=sum(x1.*exp(-1i*w(k)*n));
end
for k=1:length(w)
    X2(k)=sum(x2.*exp(-1i*w(k)*n));
end
for k=1:length(w)
    X3(k)=sum(x3.*exp(-1i*w(k)*n));
end
figure;
subplot(2,1,1);
plot(w, abs(X1));
title('幅频');
xlabel('\omega');
ylabel('|X(e^{j\omega})|');

subplot(2,1,2);
plot(w, angle(X1));
title('相频');
xlabel('\omega');
ylabel('Phase');

figure;
subplot(2,1,1);
plot(w, abs(X2));
title('幅频');
xlabel('\omega');
ylabel('|X(e^{j\omega})|');

subplot(2,1,2);
plot(w, angle(X2));
title('相频');
xlabel('\omega');
ylabel('Phase');

figure;
subplot(2,1,1);
plot(w, abs(X3));
title('幅频');
xlabel('\omega');
ylabel('|X(e^{j\omega})|');
```

```
subplot(2,1,2);
plot(w, angle(X3));
title('相频');
xlabel('\omega');
ylabel('Phase');
```

对于第二问，给a赋值0.3，w0赋值1。输出为：





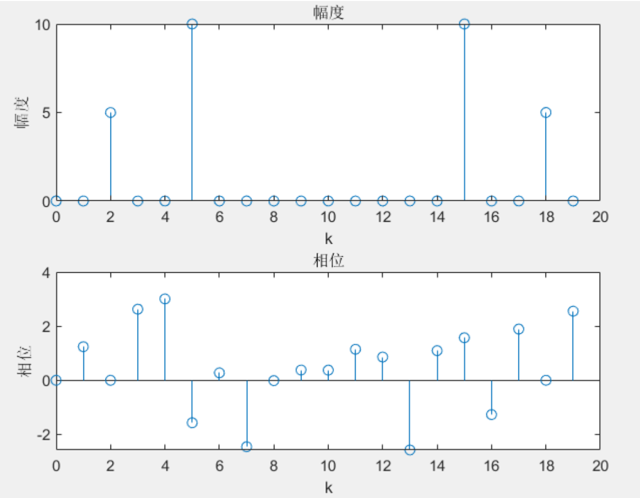


25，求序列的DFS

```
A=10;
B=20;
N=20;
n=0:19;
```

```
x=A*cos(200*pi*n/1000)+B*sin(500*pi*n/1000);
X=zeros(1,N);
for k=0:N-1
    X(k+1)=(1/N)*sum(x.*exp(-1i*2*pi*k*n/N));
end
disp(X);
figure;
subplot(2,1,1);
stem(0:N-1, abs(X));
title('幅度');
xlabel('k');
ylabel('幅度');

subplot(2,1,2);
stem(0:N-1, angle(X));
title('相位');
xlabel('k');
ylabel('相位');
```

此处赋值A为10，B为20。输出为



答案为

A/2 k=2 or 18
B/2 k=5 or 15
0 else